

PROJECT REPORT

Comparative Analysis of Human Insulin Gene using BLAST

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AIM

To analyze the similarity of the human insulin gene with sequences from other organisms using the BLAST tool and to study its evolutionary conservation.

INTRODUCTION

Insulin is an important hormone responsible for regulating blood glucose levels in the human body. It helps or allows body cells to absorb glucose from the blood for energy and regulating and to perform metabolic processes. After comparing the insulin gene sequence across different organisms, we can understand evolutionary relationships and functional conservation.

Bioinformatics tools like BLAST (Basic Local Alignment Search Tool) allow researchers to compare biological sequences and identify similarities between them.

Materials and Tools

- Computer with internet access
- NCBI database
- BLAST (Nucleotide BLAST)

Principle

BLAST is used to compare a DNA sequence with other sequences present in the database to find similarities. The results provide information such as:

- Percentage identity (similarity level)
- Query coverage
- E-value (statistical significance)

Higher similarity indicates closer evolutionary relationships.

Procedure

1. The human insulin gene sequence was downloaded from the NCBI database in FASTA format.
2. The sequence was copied and submitted to the BLAST (blastn) tool.
3. Default parameters were set as it is and used for sequence comparison.
4. The BLAST search was executed.
5. The top matching sequences and organisms were recorded and analyzed.

Results

The BLAST analysis showed high similarity of the human insulin gene with the following organisms:

- *Pongo abelii* (Sumatran orangutan)
- *Gorilla gorilla gorilla* (Gorilla)
- *Pan troglodytes* (Chimpanzee)
- *Macaca mulatta* (Rhesus macaque)
- Other primates

The percentage identity was highest among closely related primates and slightly lower in more distant species.

Discussion

The results indicate that the insulin gene is highly conserved among primates. The high similarity between human and other primate sequences shows a close evolutionary relationship.

The absence of unrelated organisms (such as plants or bacteria) in top matches confirms that the insulin gene is specific to animals and plays a crucial biological role.

This conservation highlights the importance of insulin in maintaining essential physiological functions across species.

Conclusion

From this study, it can be concluded that the human insulin gene is very similar in other primates, showing its importance and conservation., confirming its evolutionary conservation and biological significance. The study demonstrates how bioinformatics tools like BLAST can be used to analyze genetic relationships between organisms.

References

1. NCBI (National Center for Biotechnology Information)
2. BLAST (Basic Local Alignment Search Tool)